Noah Riego

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SUMMARY

Applied Physics student in the Honors College conducting independent research in nonlinear dynamics and chaotic systems, with a sole-author manuscript in preparation on reservoir-computing extrapolation of the Lorenz bifurcation diagram. Experienced in stochastic-process modeling, statistical inference, and bit-for-bit reproducible scientific computing in Python.

EDUCATION

University of Arizona - W.A. Franke Honors College
B.S. Applied Physics, Minor in Statistics and Data Science *Expected May 2028*
Physics Coursework: Intro Mechanics, Intro E&M (in progress), Intro Optics & Thermodynamics (in progress), all Honors Math Coursework: Differential Equations, Vector Calculus, Linear Algebra (in progress)

RESEARCH EXPERIENCE

Reservoir Computing for Bifurcation Extrapolation (Lorenz system) **Independent Research**
Sole-author manuscript in preparation for the Journal of Undergraduate Research in Physics and Astronomy

- Designed a controlled sweep, not previously reported, of how well echo state networks extrapolate the Lorenz system's chaotic bifurcation diagram across training range, sample density, and window position near the subcritical Hopf point.
- Held architecture and data fixed (120,000 time steps; 32 reservoir realizations per cell): extrapolation reach saturates at $\Delta\rho \approx 0.30$ beyond three training samples, and every window places the predicted collapse at attractor-coexistence onset ($\rho \approx 24.10$) rather than at the Hopf point.
- Engineered bit-for-bit reproducibility: figures regenerate from a pinned environment; all randomness traces to one master seed.

Undergraduate Research Assistant **Feb 2025 – May 2026**
Finding Research Made Easy, Vertically Integrated Projects (VIP), University of Arizona

- Built the evaluation testbed and data-collection pipeline for a faculty-supervised study of search-assisted research matchmaking; co-author on experimental design and quantitative methodology.

PROJECTS

ENSO Climate Data Pipeline
1st place, 2026 UA Data Visualization Challenge

- Built an ETL pipeline integrating five federal climate and hydrology data sources (NOAA, NCEP/NCAR, USGS, NASA) into a 7-table star-schema PostgreSQL warehouse, powering a D3.js visualization of 75 years of El Niño/La Niña events against western-U.S. river discharge.

Statistical Arbitrage Research Framework (ou-statarb)
Python, sole author | github.com/NDAR123909/ou-statarb

- Built a stochastic time-series inference library: exact AR(1) estimation of Ornstein–Uhlenbeck (mean-reverting) processes, Kalman filtering of time-varying coefficients, Johansen cointegration, and a cost-aware event-driven backtester.
- Quantified multiple-comparisons bias: a naive 465-pair scan flags 74 pairs as cointegrated where ~23 are expected by chance; applied Benjamini–Hochberg FDR control plus stability, half-life, and Hurst-exponent filters, which reject ~97% of candidate pair-folds.
- Validated out of sample across 19 walk-forward folds on 31 DJIA stocks (2006–2017), reporting the median fold rather than the best backtest; running a preregistered 6-month paper-trading experiment with rules frozen in a git-timestamped commit before day one.
- Powers my agent in the LTP Liquidity Arena 2026, a global live-capital algorithmic trading competition: 6th of 30 advancing teams in Phase 1 and a finalist in the live-mainnet final (Sep–Nov 2026).

EXPERIENCE

Founding Engineer (part-time) **Nov 2025 – Sep 2026**
Lynnapse

- Ran a pre-registered A/B study with metric and thresholds locked before any run; traced an apparent +22.5% dev-set uplift to false positives via a URL audit, confirmed 0% real uplift on held-out data, and kept the LLM prioritization layer off by default.
- Built the LLM inference layer behind a queue-backed FastAPI service with retry and pacing; across 13 test crawls in three batches, held cost to ~$0.0015 per crawl (10x under budget) and absorbed 13–20% LLM call failures with zero crawl failures.

SKILLS

Physics & Mathematics: Nonlinear dynamics and chaos (bifurcation analysis, Lorenz system), stochastic processes (Ornstein–Uhlenbeck/AR, Kalman filtering), reservoir computing
Data Analysis: Statistical inference, FDR multiple-testing control, Monte Carlo simulation, bootstrap inference, time-series analysis (cointegration, unit-root tests), walk-forward validation
Computing: Python (NumPy, SciPy, pandas, statsmodels, scikit-learn, Matplotlib), C++, C, Linux/Bash, LaTeX, Git, GitHub Actions CI, SQL/PostgreSQL, JavaScript/D3.js